News Retrieval under Vocabulary Mismatch

BM25, neural search and hybrid retrieval

Group 5 · Information Retrieval
Faculty mentor: Abhinav Kumar
Parth Raghuvanshi — 2023A7PS0574P
Sushruth Reddy Challa — 2023A7PS0504P
Hamsini G — 2023A7PS0006P
Kritika Pandey — 2023A7PS0022P

1. Primary objective

Find out when a news search engine benefits from semantic matching and when exact wording still matters. We compare BM25 with E5 dense retrieval over the same articles, focusing on vocabulary mismatch, negation, ambiguous words and event dates. The aim is a working comparison demo and an evidence-based account of their strengths and failures. E5 is a neural bi-encoder; the findings concern this model and setup, rather than all LLMs.

2. Dataset and evaluation criteria

Selected dataset: TREC-News mirror — liuqi6777/trec-news. The downloaded corpus has **594,977 records**. Excluding 388 records with no title or body leaves **594,589 articles** for both engines. Title-only articles remain in the baseline so the two systems search the same IDs.

We retain the background-linking and custom diagnostics, and add a paired ad-hoc news experiment. The **57 original background-linking topics** have **15,655 supplied relevance judgments**. Our **40 custom queries** have ten questions per category: vocabulary mismatch, negation, homonyms and temporal constraints. All 1,444 top-20 pooled pairs are assessed; the initial 737 decisions remain unchanged.

Integrated robustness study: 50 NIST Core 2018 topics are now evaluated on the available news corpus, using original titles, official descriptions and one provisional rewrite per topic (150 formulations). The 26,233 source qrel pairs match NIST. Seven judged articles are absent and eleven are empty; one unavailable pair is positive. All original qrels remain in primary denominators. This is an available-corpus adaptation, not an official WaPo v2 reproduction. Hamsini's 50-topic TREC-COVID study is preserved as a separate cross-domain experiment.

Success measures: nDCG@5 measures how well the first five results are ordered by relevance; MAP@1000 measures recovery and ordering of known relevant articles across the saved top 1,000. We also report Precision@5, category means, judged coverage and paired query-bootstrap intervals. AP divides by all known positives for the query. Main nDCG gains equal relevance grades: 0–4 for the 57-topic reference and 0–2 for Core 2018, COVID and custom queries.

Reference status: custom results use evidence-backed full-article **AI judgments** under a 0–2 rubric. **Human review is planned next and has not yet been completed.** New news rewrites are AI-authored, frozen before retrieval, and await independent human intent review; their scores reuse human NIST labels conditionally on preserving the narrative. Official descriptions change length/detail, so they are a query-form diagnostic. COVID rewrites retain their source study's single-annotator verification. Custom MAP is shallow-pool; none of these labels establish that unjudged articles are irrelevant.

3. Related research: approach, findings and relevance

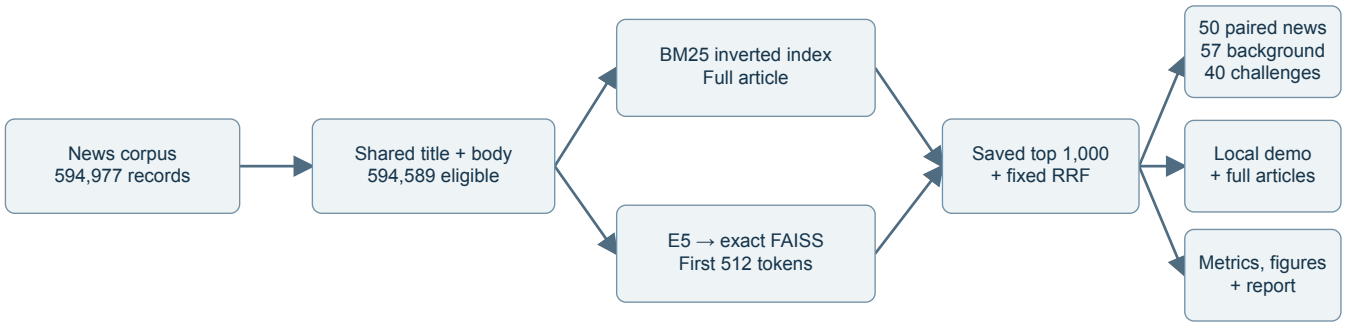
Thakur et al. (2021), BEIR: A Heterogenous Benchmark for Zero-shot Evaluation of Information Retrieval Models. The paper evaluates ten retrieval systems across 18 datasets. BM25 is a robust baseline; reranking and late-interaction methods have strong average quality at greater computational cost. This supports keeping BM25 as a serious comparator and examining different query types.

Wang et al. (2022; revised 2024), Text Embeddings by Weakly-Supervised Contrastive Pre-training. E5 learns text embeddings through contrastive training on text pairs and reports strong transfer on BEIR/MTEB tasks. We use a released E5 checkpoint to test whether this semantic representation helps with news queries whose wording differs from relevant articles.

Weller, Lawrie and Van Durme (2024), NevIR: Negation in Neural Information Retrieval. The authors compare documents differing in negation. Many retrievers struggle with polarity; cross-encoders perform better than bi-encoders. This motivates explicit negation tests instead of assuming that semantic similarity enforces a negative constraint.

Cormack, Clarke and Büttcher (2009), Reciprocal Rank Fusion outperforms Condorcet and individual Rank Learning Methods. The paper combines ranked lists through reciprocal-rank contributions and reports gains over individual systems in its experiments. We test fixed RRF with k=60 as an extension, without fitting weights to our judgments.

4. Approach and architectural reasoning



Component	Method and reason for the choice
BM25	Lucene inverted index; k1=0.9, b=0.4. A strong term-based baseline makes exact-name and wording effects visible.
Dense retrieval	E5-base-v2: query:/passage: prefixes, mean pooling, normalized 768-D vectors; exact FAISS cosine ranking. Tests semantic matching without approximate-index error.
Hybrid and inspection	Fixed equal-weight RRF, sum(1/(60+rank)); no tuning. Article lookup and the demo connect ranked IDs to full text and relevance evidence.

Both engines use title plus body. English lexical analysis lowercases and tokenizes, preserves stopwords and does not stem. BM25 reads full bodies; E5 retains the first **512 tokens**, including its prefix and special tokens. This difference is audited and remains a limitation of the comparison. BM25 treats “NOT” as a term, with no Boolean negation or date filter. The lexical foundations follow Manning et al., *Introduction to Information Retrieval*, Chapters 6 and 11.

5. Current findings and what they mean

All systems are implemented. Each row below is a separate 50-topic experiment; scores are not averaged across domains or formulations. The table reports **nDCG@5**, with graded gains equal to the reference labels.

Separate experiment	BM25	E5	RRF
News · original titles	0.4454	0.3878	0.4453
News · provisional rewrites	0.2868	0.4113	0.4447
News · official descriptions	0.3796	0.3736	0.4129
COVID · original keywords	0.6687	0.5746	0.7504
COVID · rewrite A	0.3269	0.4208	0.4792
COVID · rewrite B	0.2694	0.4303	0.4200

Paired news finding: E5's rewrite-minus-title change exceeds BM25's by **+0.1820 nDCG@5**, with a 95% paired interval **[+0.0784, +0.2905]** and Holm-adjusted p=**0.0168**. The direction also holds for MAP and bpref, but the condensed-nDCG@10 interval includes zero. RRF stays near its original news score, while it drops substantially in the COVID rewrites. These are separate, conditional patterns—not proof that domain alone causes the difference.

Reference sensitivity: judgment coverage changes when wording changes. Condensed nDCG removes unjudged hits; bpref supplies another incomplete-reference view. Neither supplies missing judgments. The news rewrite conclusions remain provisional until independent intent review; title-versus-description differences also include query length and specificity.

Preserved diagnostics: on the original 57 background-linking topics, BM25/E5/RRF nDCG@5 remains **0.3975/0.4576/0.4900**. On custom40, BM25/E5 is **0.5133/0.4362** and pooled MAP is **0.4428/0.4072**; family intervals include zero. Initial20 nDCG@5 was **0.4621/0.4897**. A separate post-hoc empty-body filter raises E5 to **0.5492** within the initial judged pool; it is not a cleaned-corpus replication.

Resources: the news BM25/dense indexes occupy 367.6 MiB/1.72 GiB; E5 weights add about 418 MiB. Existing T4 corpus encoding took about 5 h 43 min and is reused. New query batching records amortized throughput, not single-query latency. Full-body BM25 versus E5's first 512 tokens remains an input-coverage confound. Historical evaluation files and source-study results are preserved; current checks and exact values are in the reproducibility package.

6. Key innovations and stretch objectives

The project combines controlled query rewording with constraint diagnostics, source-linked relevance evidence and an audit of E5's retained input. Separate news and COVID experiments test whether the same robustness pattern holds in different settings.

Completed stretch objectives: fixed RRF on the original 57 topics; exact-content deduplication and empty-body filtering within judged candidates; gain/confidence sensitivity; and a full-article demo with an E5 input toggle. **Further work:** independently review news rewrite intent and custom labels, then test controlled polarity pairs, equal-input retrieval and a cross-encoder. These extensions have no claimed results.

Expanded evaluation completed: 40 queries, ten per category; all 1,444 designated pairs assessed. Initial, added and combined results are retained separately, with topic-family uncertainty and full-text evidence.

7. Key milestones and tentative timeline

Completed dates below follow saved artifact timestamps in India time. Remaining durations are planning estimates, not a new submission deadline.

Milestone	Timing and status
Shared corpus and BM25 index	19 Oct 2026 · completed.
Full E5 index and original 57-topic evaluation	24 Oct 2026 · completed.
Hybrid, 40-query study, demo and report	25 Oct 2026 · completed; integrated news/COVID experiments added 2 Nov.
Begin human review and member checks	Next session: 2–3 hours for an initial sample; tentative.
Public repository and demo rehearsal	Repository published; allow about 1 hour for rehearsal.
Final link, page-count and submission checks	Allow 30 minutes before the course deadline; one member submits.

8. Contribution matrix and delivery

Member	Assigned ownership	Completed team component
Parth Raghuvanshi	data/BM25	Corpus audit, preprocessing, BM25 index and GPU handoff.
Sushruth Reddy Challa	E5/FAISS	E5 encoder, GPU workflow, vector index and dense search.
Hamsini G	evaluation	COVID rewrite experiments, query tooling and paired statistics.
Kritika Pandey	demo/report	Comparison demo, figures, failure analysis and report package.

Rows distinguish assigned responsibility from verified personal work; Hamsini's source study is credited.

Working demo: run `.venv/bin/python -m news_ir.demo` and open `http://127.0.0.1:8765`. Saved views show scores and evidence; new live queries are explicitly unjudged. No additional GPU job is needed. GitHub repository

Submission file: `5_NewsIR_BM25_Dense.pdf`. Only one group member submits this single four-page PDF.